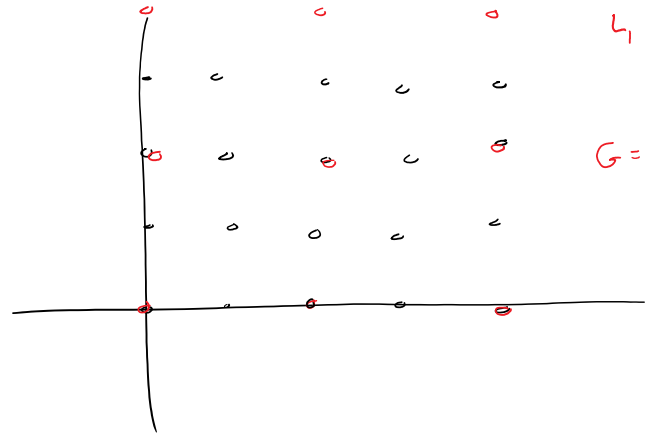


$L_1 \subset L_2$

$F = [b_1 \ b_2 \ \cdots \ b_n]$

$G = [g_1 \ g_2 \ \cdots \ g_n]$

$g_1 = F x_1$

$g_2 = F x_2$

$\vdots$

$g_n = F x_n$

$[g_1 \ g_2 \ \cdots \ g_n] = F \underbrace{[x_1 \ x_2 \ \cdots \ x_n]}_{X}$

$G = F X.$

$\underline{\det(G)} = \underline{\det(F)} \underbrace{\det(X)}_{\text{integer.}}$

$\det(F) \mid \det(G)$

$\begin{cases} G : L_1 \\ F : L_1 \end{cases}$

$G = F X_1$

$F = G X_2$

$\det(G) \mid \det(F)$

$\det(F) \mid \det(G)$

$x(t) \approx \sum_{i=1}^{n} c_i p_i(t)$

$X = \sum_{i=1}^{n} c_i P_i$

~~suppose $\{p_i\}$ are~~

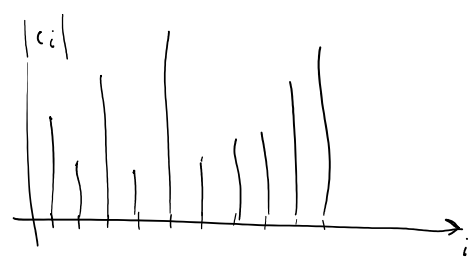
orthonormal,

$\boxed{c_i = \langle x(t), p_i(t) \rangle}$

Ex: $\underbrace{p_i(n) = e^{+jni\frac{2\pi}{N}}}$

$x(n) = \sum_{i=0}^{n-1} \underbrace{c_i} e^{+jni\, 2\pi/N}$

$c_i = \dfrac{\langle x(n), p_i(n) \rangle}{\langle p_i(n), p_i(n) \rangle}$



Error: $\left\| x(t) - \sum_{i=1}^{n} c_i p_i(t) \right\|_2^2$

$$= \left\langle x(t) - \sum_{i=1}^{n} c_i p_i(t),\ \underbrace{x(t) - \sum_{i=1}^{n} c_i p_i(t)}_{e(t)} \right\rangle$$

$$= \left\langle x(t) - \sum_{i=1}^{n} c_i p_i(t),\ x(t) \right\rangle - \underbrace{\left\langle x(t) - \sum_{i=1}^{n} c_i p_i(t),\ \sum_{i=1}^{n} c_i p_i(t) \right\rangle}_{0}$$

$$= \langle x(t), x(t) \rangle - \left\langle \sum_{i=1}^{n} c_i p_i(t),\ x(t) \right\rangle$$

$$= \langle x(t), x(t) \rangle - \left\langle \sum_{i=1}^{n} \langle x(t), p_i(t) \rangle p_i(t),\ x(t) \right\rangle$$

$$= \langle x(t), x(t) \rangle - \sum_{i=1}^{n} \underbrace{\langle x(t), p_i(t) \rangle}_{c_i} \underbrace{\langle p_i(t), x(t) \rangle}_{c_i^*}$$

$$= \|x(t)\|^2 - \sum_{i=1}^{n} |c_i|^2 \geq 0$$

$$\boxed{\|x(t)\|^2 \geq \sum_{i=1}^{n} |c_i|^2}$$ Bessel's inequality.

If $\{p_i(t)\}$ is complete, $x(t) = \sum_{i=1}^{n} c_i p_i(t)$

$$\left\| x(t) - \sum_{i=1}^{n} c_i p_i(t) \right\|^2 = 0$$

$$\Rightarrow \boxed{\|x(t)\|^2 = \sum_{i=1}^{n} |c_i|^2}$$ Parseval's thm.

$$x(t) = \sum_{i=1}^{n} c_i p_i(t)$$

$$y(t) = \sum_{i=1}^{n} d_i p_i(t)$$

$$\boxed{\begin{aligned} \langle x(t), y(t) \rangle &= \sum_{i=1}^{n} c_i d_i^* \\ &= \langle \underline{c}, \underline{d} \rangle \end{aligned}}$$ Parseval's thm.